

Multimedia Computing

By

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Content

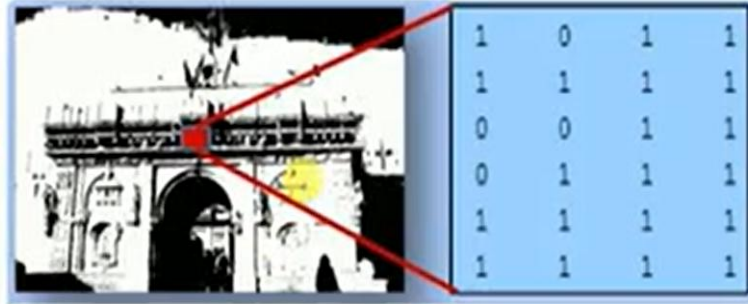
- Image types
- LOOK-UP Tables
- Contrast Stretching
- Intensity Transformation
- Histogram Equalization
- Histogram Specification
- Arithmetic Operations
- Logic Operations

Types of image

- ▶ Binary image
- ▶ Gray scale image
- ▶ Color image

Types of image

- Binary Image: Each Pixel is just Black and white, i.e. 0 or 1
- Ex: < 462x512 logical >



- Gray Scale Image: Each Pixel is shade of Gray, its 0(black) to white(255)
- i.e. each pixel-8 bits
- Ex : <462x512 unit8>



- Color Image or RGB Image, Each pixel corresponds to 3 values.
- Combination of Red, Green and Blue
- <462x512x3 unit8>



RGB image

An RGB image is essentially three images layered on top of one another; Red , green, blue each pixel in them being 8 bits(intensity value ranges 0-255). To store a single pixel of an RGB need to store 8 bits for all three colors total 24bit per pixel

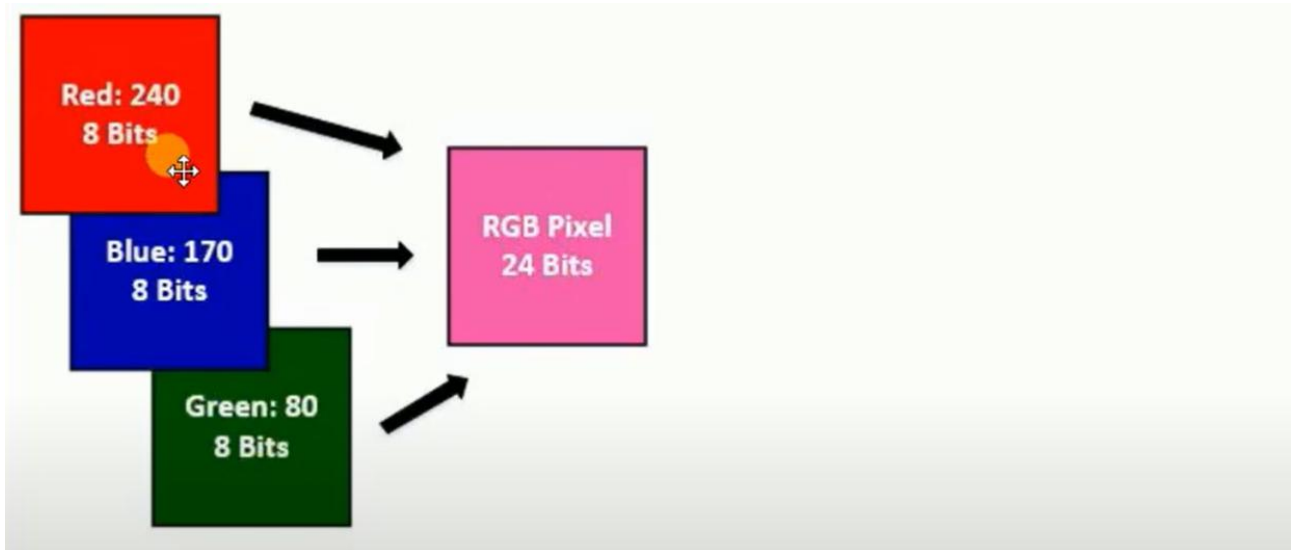


Image Data type

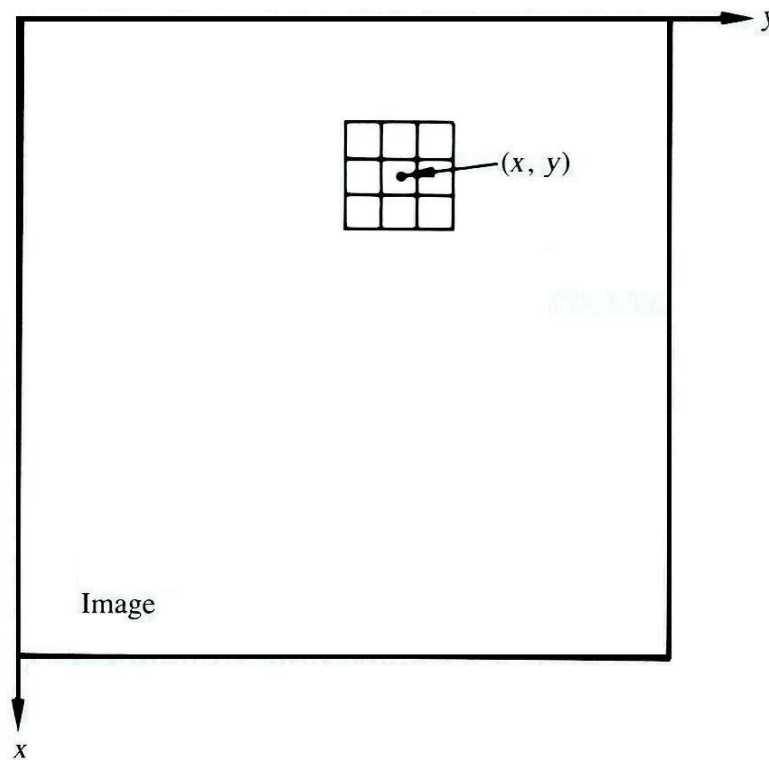
- int8-8 bit integer, Range -128 to 127
- unit8-8 bit unsigned integer, Range 0 to 255
- int16-16 bit integer, Range -32768 to 32767
- uint16-16 bit unsigned integer, Range 0 to 65535
- double-Double precision real number, Machine Specific

Point processes

- ▶ Point processes :
 - ▶ The simplest of the basic image processing operations
 - ▶ Based solely in the pixel's value
 - ▶ Easily implemented as look-up tables
- ▶ The model of spatial domain methods
 - ▶ Spatial domain point processing: operate directly in the aggregate of pixels.

Spatial domain methods

- The model for image processing
- Let T be an operator in f defined over some neighborhood of (x,y) or a set of input image
- $g(x,y) = T[f(x,y)]$
 - $g(x,y)$: processed image
 - $f(x,y)$: input image



Look-up Table (LUT)

An LUT is simply a table of cross-references linking *index numbers* to *output values*. The most common use is to determine the colors and intensity values with which a particular image will be displayed, and in this context the LUT is often called simply a *colormap*.

The idea behind the colormap is that instead of storing a definite color for each pixel in an image, for instance in 24-bit RGB format, *each pixel's value is instead treated as an index number into the colormap*. When the image is to be displayed or otherwise processed, the colormap is used to look up the actual colors corresponding to each index number. Typically, the output values stored in the LUT would be RGB color values.

There are two main advantages to doing things this way. *Firstly, the index number can be made to use fewer bits than the output value in order to save storage space*. For instance an 8-bit index number can be used to look up a 24-bit RGB color value in the LUT. Since only the 8-bit index number needs to be stored for each pixel, *8-bit color images take up less space than a full 24-bit image of the same size*.

Look-up Table (LUT)

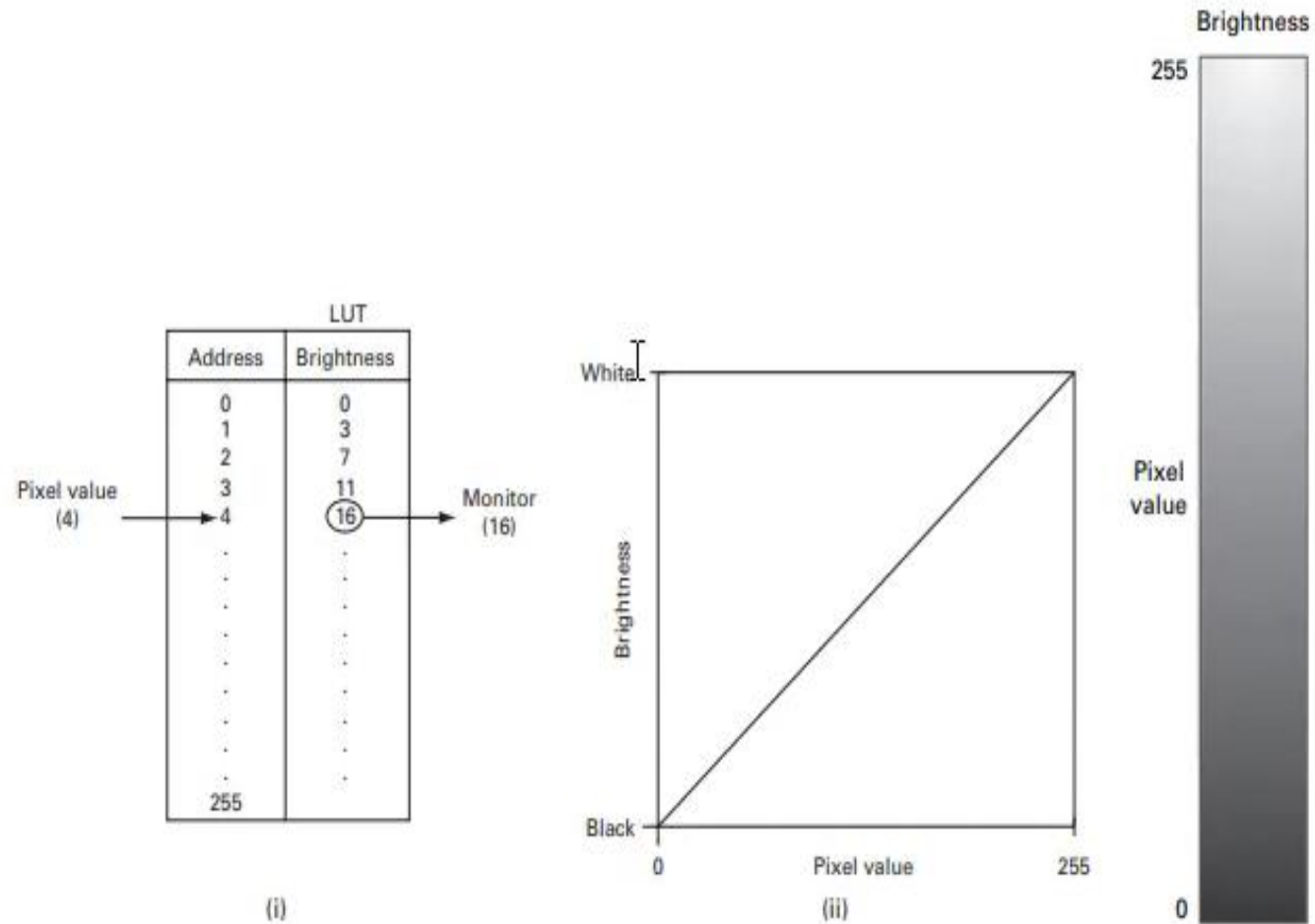
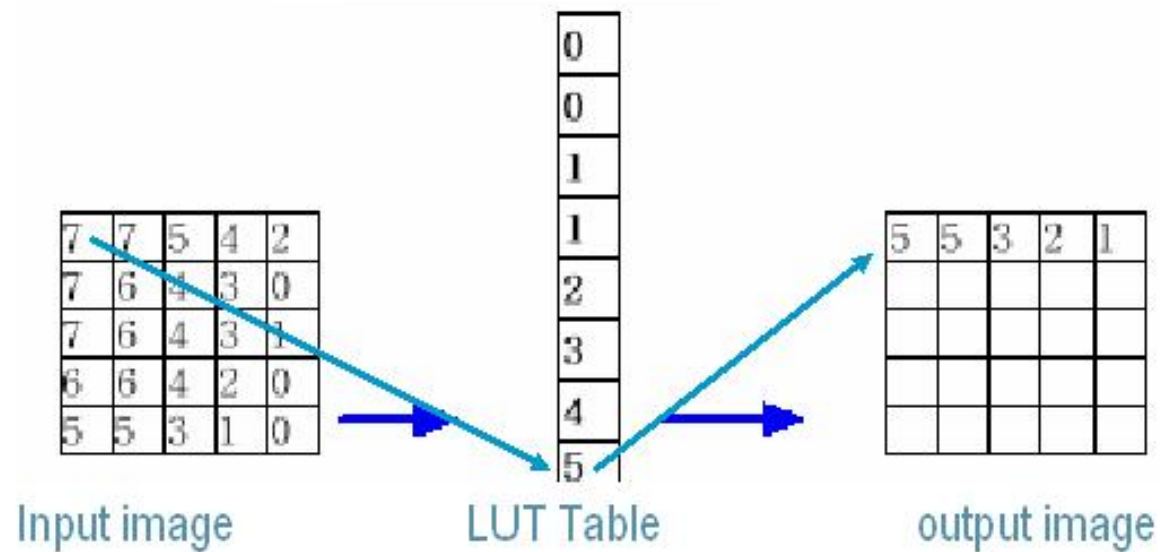


Figure 5.9 The look-up table as (i) a table stored in memory, (ii) a graph connecting pixel values to brightness levels and (iii) a distribution of gray values. (In this case, (ii) and (iii) represent the identity or default look-up table.)

Look-up Table (LUT)

▶ LOOK-UP Tables

- ▶ use the current pixel value as the array index → simple arrays
- ▶ new value is the array element pointed by this index



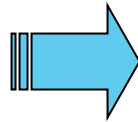
Operation of a 3-bit look-up table.

Look-up Table (LUT)

► LOOK-UP Tables



Original image



Output image after
LUT

The model of point processing

- ▶ Point processing: the simplest form of T
 - ▶ 1 x 1 neighbor
 - ▶ g depends on only on the value of f at (x,y)
 - ▶ Gray-level transformation(or mapping)
 - ▶ $s = T(r)$
 - ▶ $s(\text{processed image})$: the gray level of the $g(x,y)$
 - ▶ $r(\text{input image})$: the gray level of the $f(x,y)$

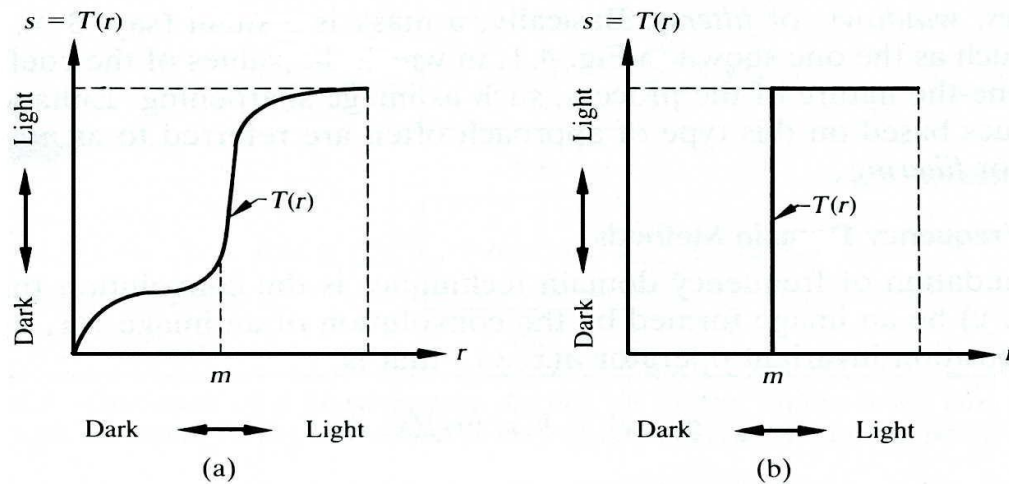
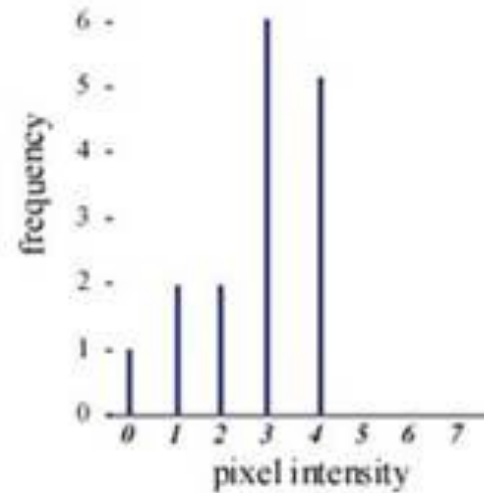


Fig. Gray-level transformation for contrast enhancement [Gonzalez]: (a) the effect is contrast stretching, and (b) binarization.

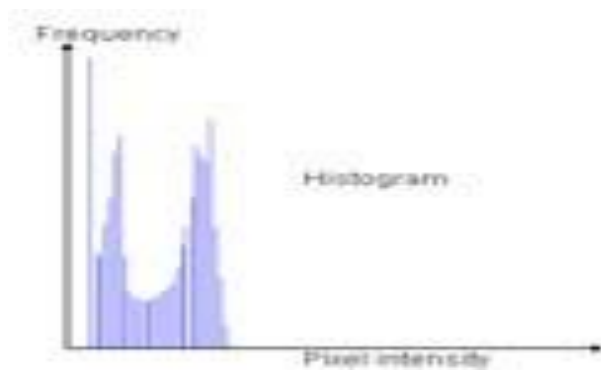
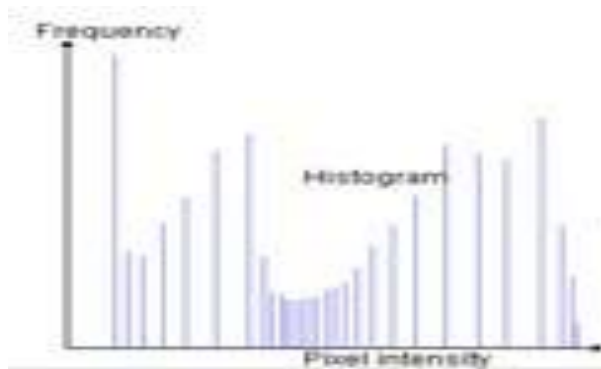
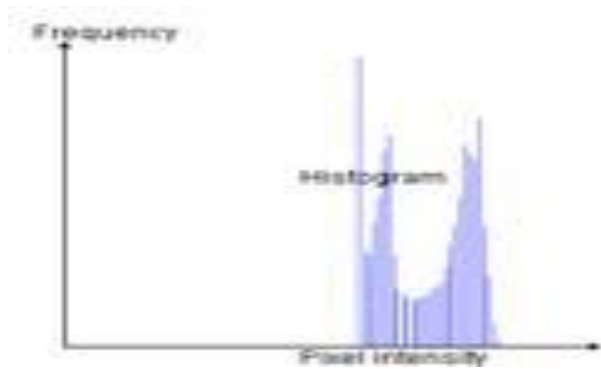
Histogram

- ▶ Histogram
 - ▶ A simple bar graph that stands for pixel intensities.
 - ▶ The pixel intensities are plotted along the x-axis and the number of occurrences for each intensity are plotted along the y-axis.
 - ▶ Provide information about contrast and overall intensity distribution of an image

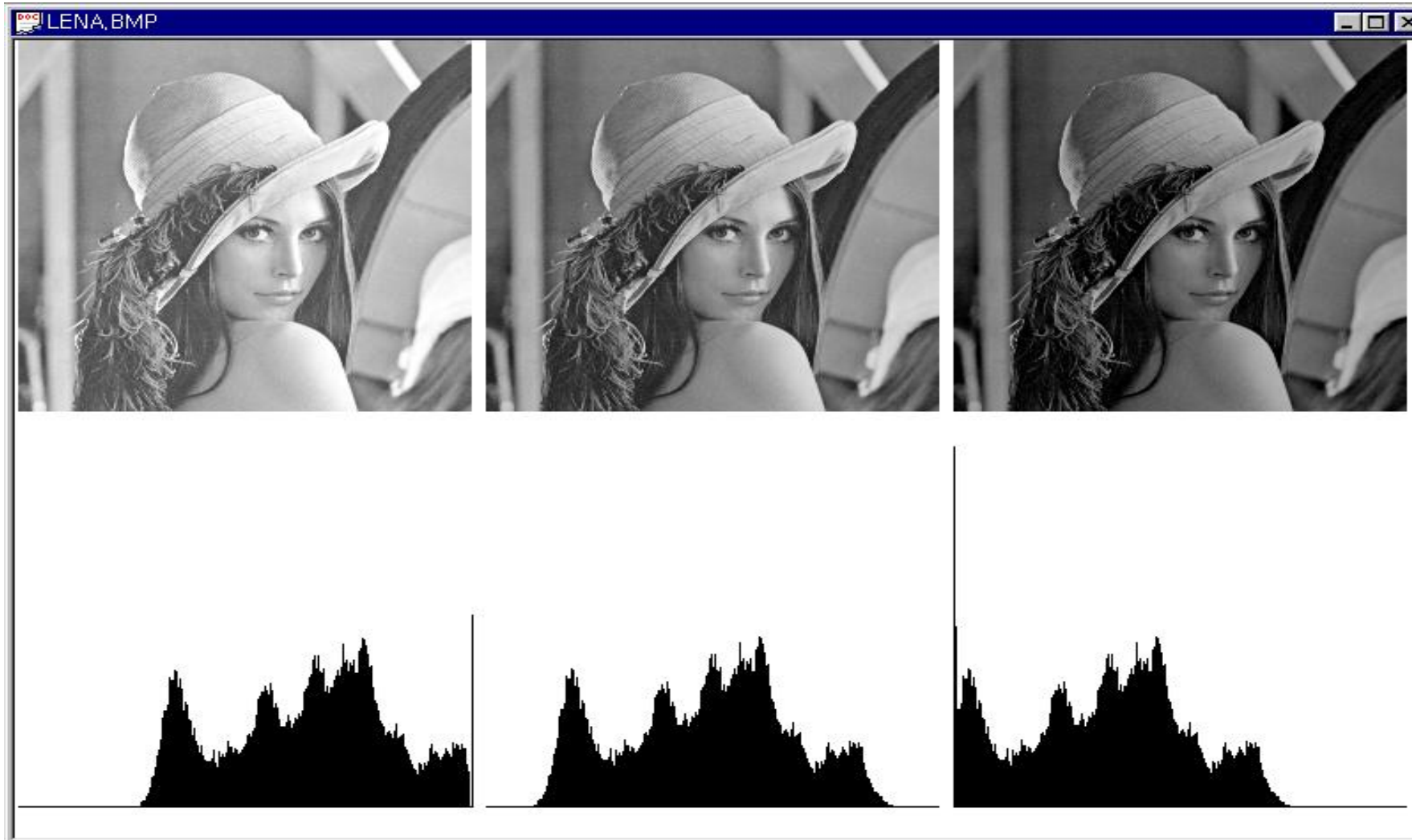
4	4	3	3
4	4	3	3
4	1	2	3
0	1	2	3



Histogram



Histogram

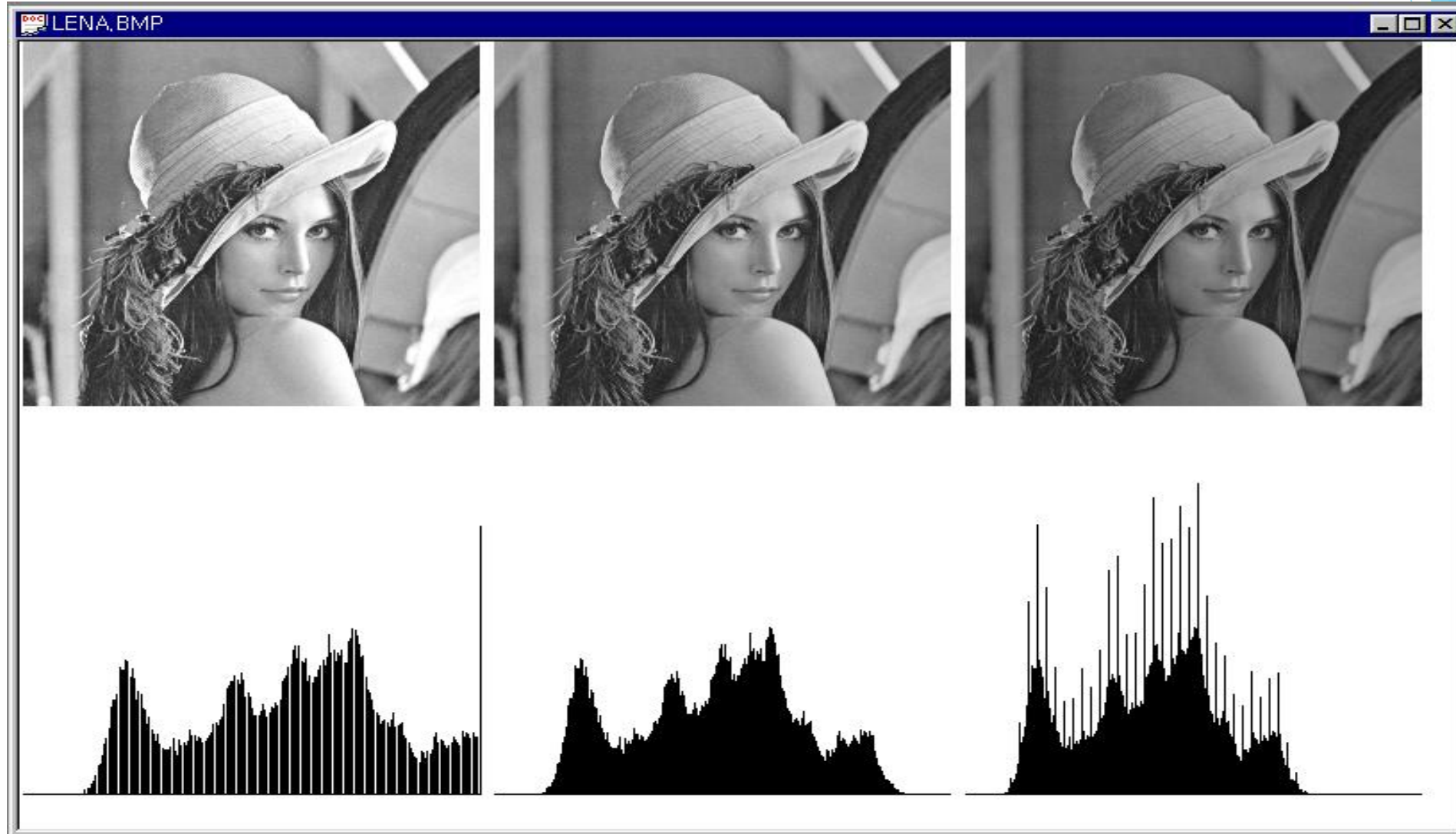


Original Image + 40

Original Image

Original Image - 40

Histogram



Original Image $\times 1.2$

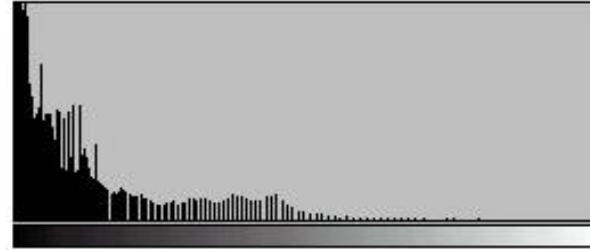
Original Image

Original Image $\div 1.2$

Histogram



image



Histogram



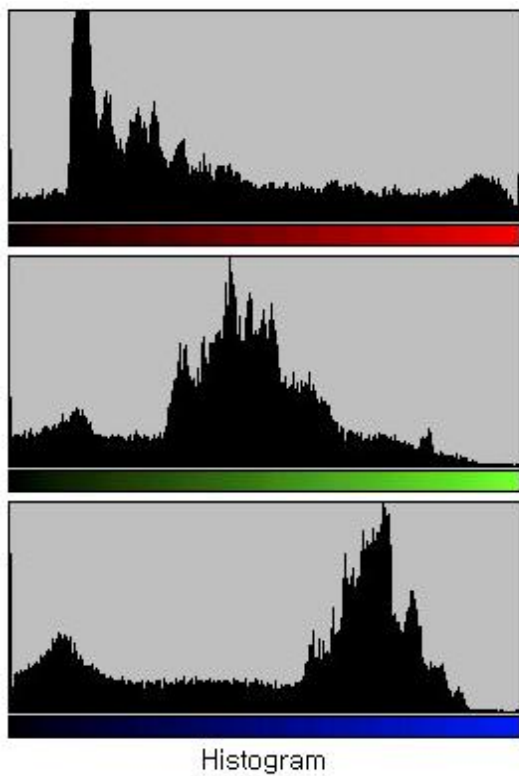
image



Histogram

Histogram

► Histogram in Gray-level and Color Image



RGB



Original image



Original image

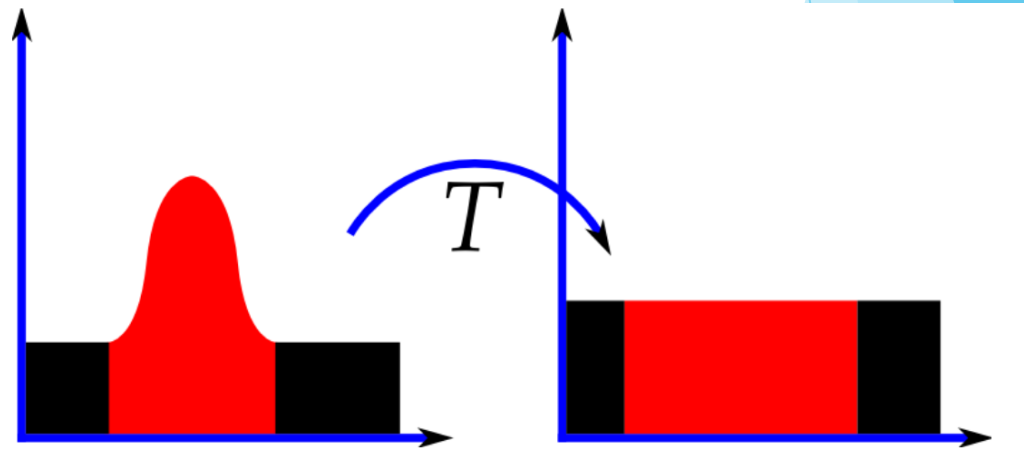


Histogram

Gray
19

Histogram Equalization

- ▶ Histogram Equalization
 - ▶ (Goal) to obtain a uniform histogram for the output image
 - ▶ Mapping of gray level r into gray level s s.t. the distribution of gray level s is uniform.
 - ▶ Spreading: the peaks and valleys will be shifted (due to approximation in digitized space)
- ▶ Histogram Equalization Steps
 1. Compute histogram.
 2. Calculate normalized sum of histogram
 3. Transform input image to output image



Histograms of an image before and after equalization.

Histogram Equalization

- ▶ (def)

- ▶ r : represent the gray level, $0 \leq r \leq 1$

- ▶ Pixel values are continuous quantities

- ▶ Discrete formulation: allow pixel values to be interval $[0, L-1]$

- ▶ Transform $s = T(r) = \int_0^r p_r(w) dw, 0 \leq r \leq 1$

- ▶ Produce a level s for every pixel value r in the original image

- ▶ The transform must satisfies the following conditions

$\in$

Condition(a): $T(r)$ is single-values and monotonically increasing in $r \in [0,1]$

Condition(b): $0 \leq T(r) \leq 1$ for $r \in [0,1]$

$\in$

- ▶ Inverse transform $r = T^{-1}(s), s \in [0,1]$

Histogram Equalization

- ▶ A gray-level transformation function

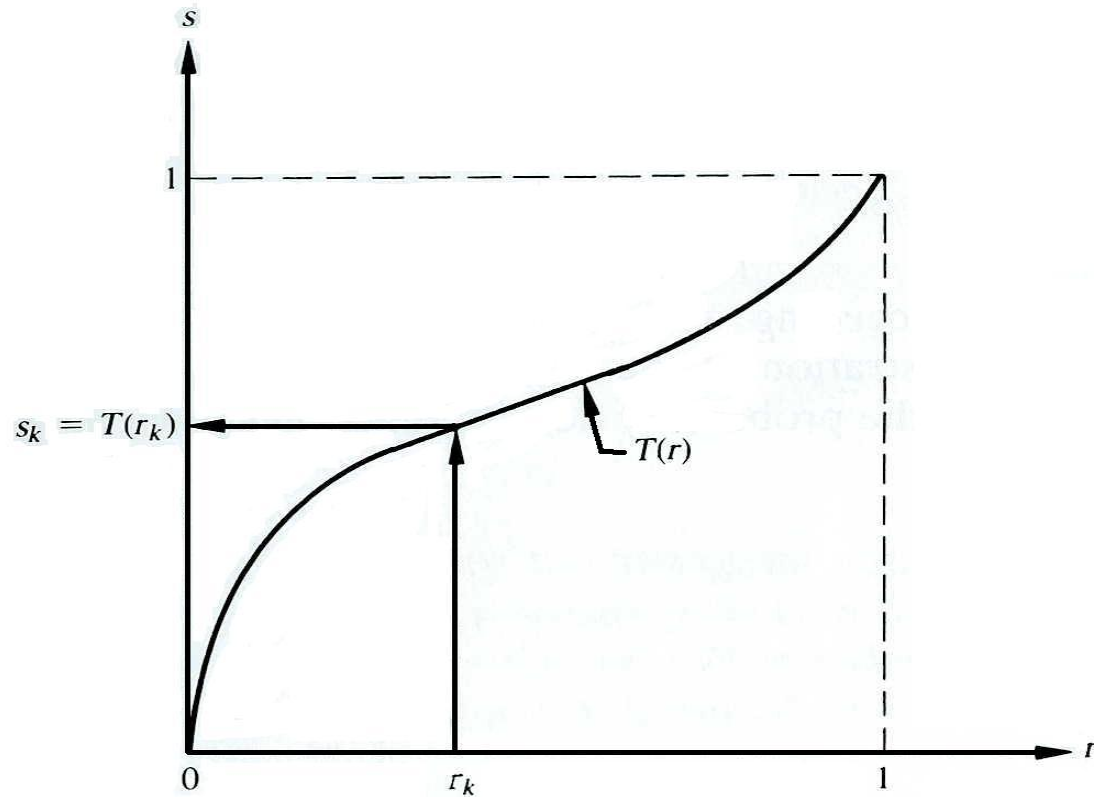


Fig. A gray-level transformation function.
[Gonzalez]

Histogram Equalization

- ▶ The gray levels may be viewed as random quantities in $[0,1]$
 - ▶ Can be characterized by probability density functions
 - ▶ Original gray levels: $p_r(r)$
 - ▶ Transformed gray levels: $p_s(s)$
- ▶ From elementary probability theory,
 - ▶ If $p_r(r)$ and $T(r)$ are known and $T^{-1}(s)$ satisfies condition (a) the probability density function of the transformed gray levels is

$$p_s(s) = \left[p_r(r) \frac{dr}{ds} \right]_{r=T^{-1}(s)}$$

Histogram Equalization

- ▶ Consider the $T(r)$ with CDF(Cumulative distribution function) of r , i.e.,

- ▶ $s = T(r) = \int_0^r p_r(w) dw, 0 \leq r \leq 1$

- ▶ Then, $\frac{ds}{dr} = p_r(r)$

- ▶ Uniform density of the transformed variables s

$$\begin{aligned} p_s(s) &= \left[p_r(r) \frac{1}{p_r(r)} \right]_{r=T^{-1}(s)} \\ &= [1]_{r=T^{-1}(s)} \\ &= 1 \end{aligned}$$

Histogram Equalization

► Uniform density transformation method

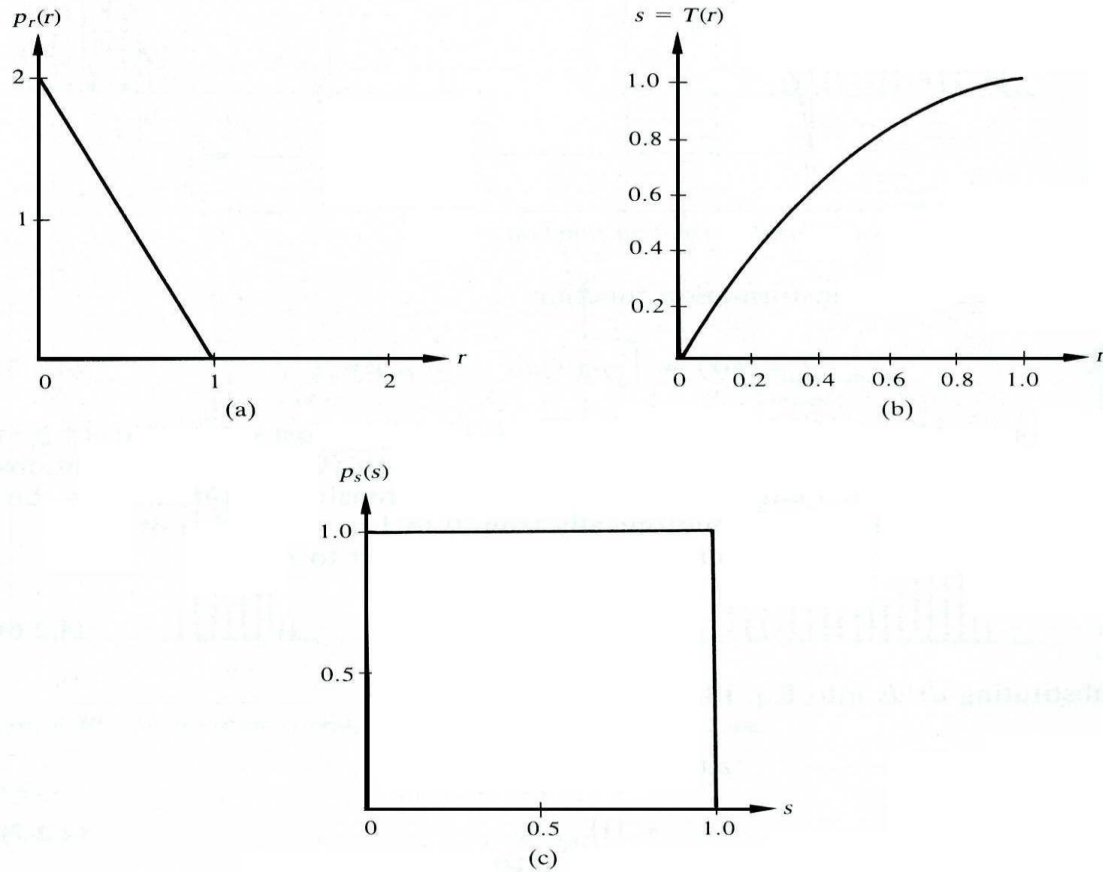


Fig. Illustration of the uniform density transformation method: (a)original probability density function; (b)transformation function; (c)resulting uniform density [Gonzalez]

Histogram Equalization

- ▶ The effects of H.E.
 - ▶ H.E. stretches contrast (expand the range of gray levels) for gray levels near histogram maxima
 - ▶ Compresses contrast in areas with gray levels near histogram minima.
 - ▶ Contrast is expanded for the most of the image pixels => H.E. usually improves the detectability of many image features.
 - ▶ The resulting histogram is not flat
 - ▶ nothing in the discrete approximation of the continuous result previously derived says that it should be flat.
 - ▶ Similar effect of enhancement could be achieved by manual contrast stretching approach

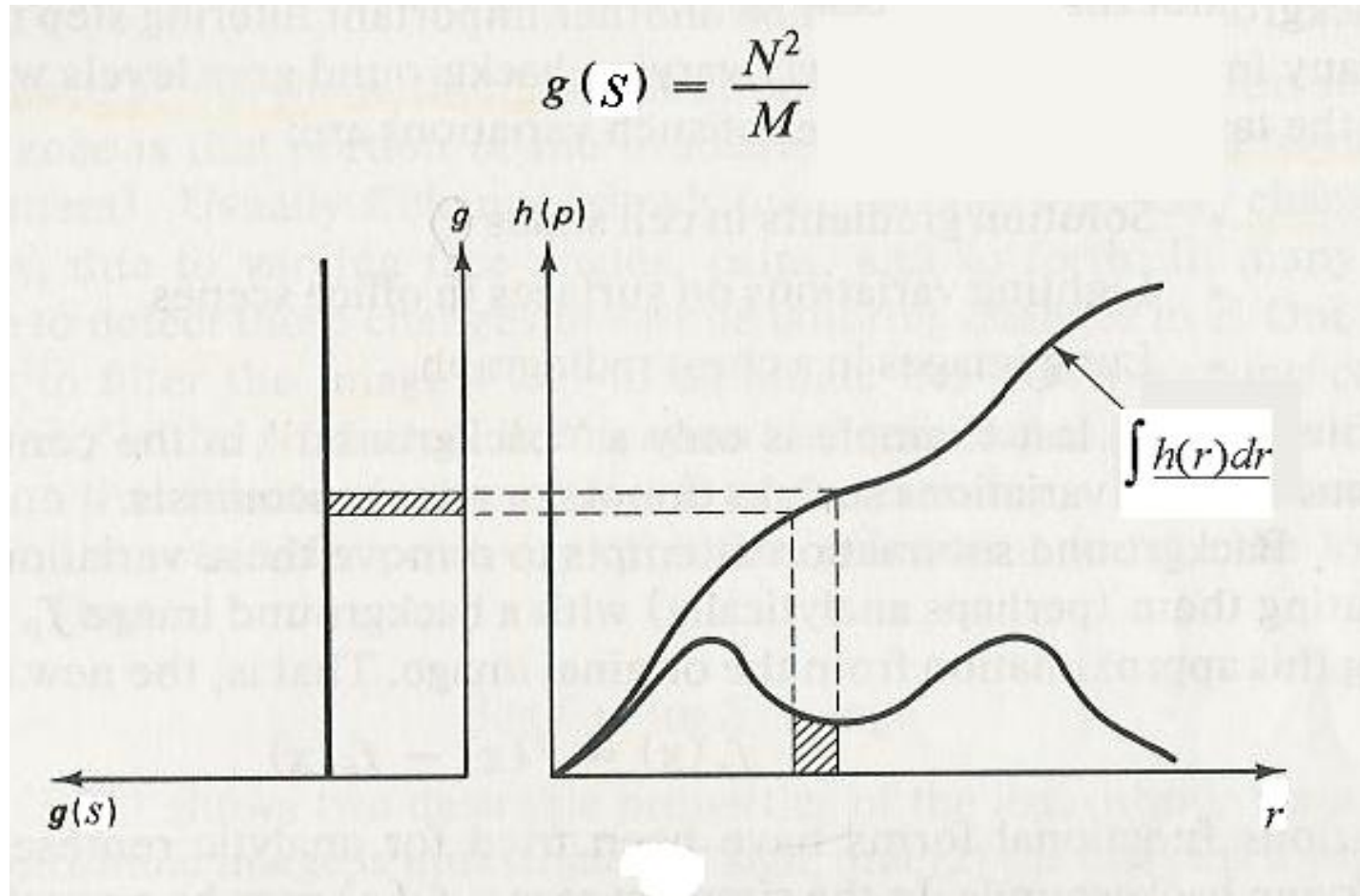
Histogram Equalization

- ▶ The mapping of H.E.
 - ▶ To map a small interval of gray level dr onto an interval ds , $\Rightarrow g(s)ds = h(r) dr$
 - ▶ $h(r)$: histogram of input image, $g(s)$: new histogram
 - ▶ $g(s)$ is to be uniform for H.E., i.e., $G(s) = N^2 / M$
 - ▶ N^2 : the number of pixels in the image
 - ▶ M : the number of gray levels
 - ▶ Then, we have the equation for the normalized cumulative histogram as follows.

- ▶ $T(r) = \frac{M}{N^2} \left(\int_0^r h(w)dw \right)$

Histogram Equalization

- Basis of H.E. [D.H. Ballard]

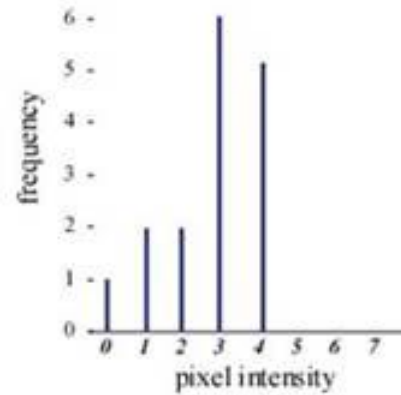


Histogram Equalization

[Method]

- Step 1: Compute histogram

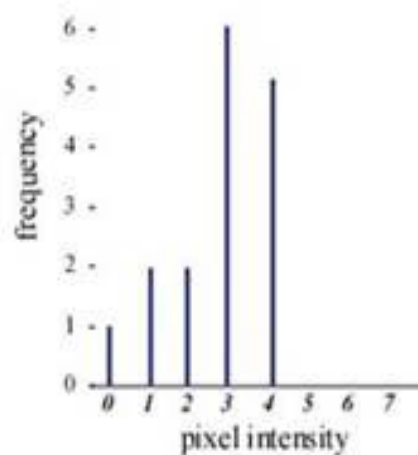
4	4	3	3
4	4	3	3
4	1	2	3
0	1	2	3



Histogram Equalization

- Step 2 : Calculate normalized sum of histogram

- $1 * (7/16) = 0.43$
- $3 * (7/16) = 1.31$
-



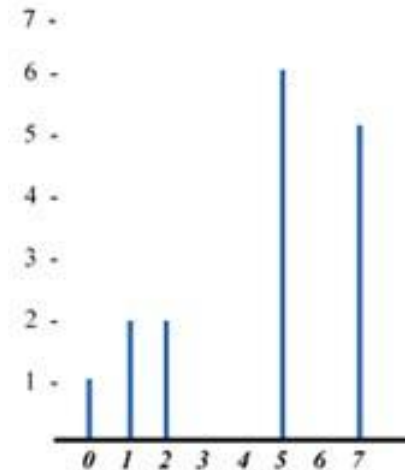
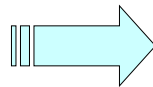
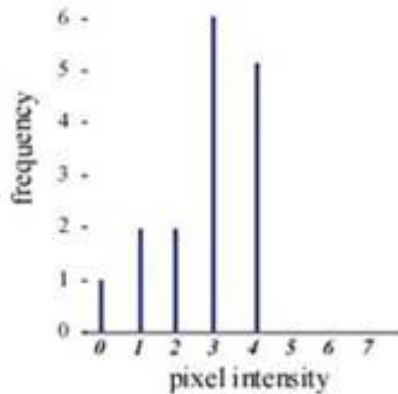
	sum	normalized sum
0	1	0.43
1	3	1.31
2	5	2.18
3	11	4.81
4	16	7.00
5	16	7.00
6	16	7.00
7	16	7.00

Histogram Equalization

- Step 3: Transform input image to output image

Image transform using LUT

4	4	3	3
4	4	3	3
4	1	2	3
0	1	2	3



7	7	5	5
7	7	5	5
7	1	2	5
0	1	2	5

Histogram Equalization

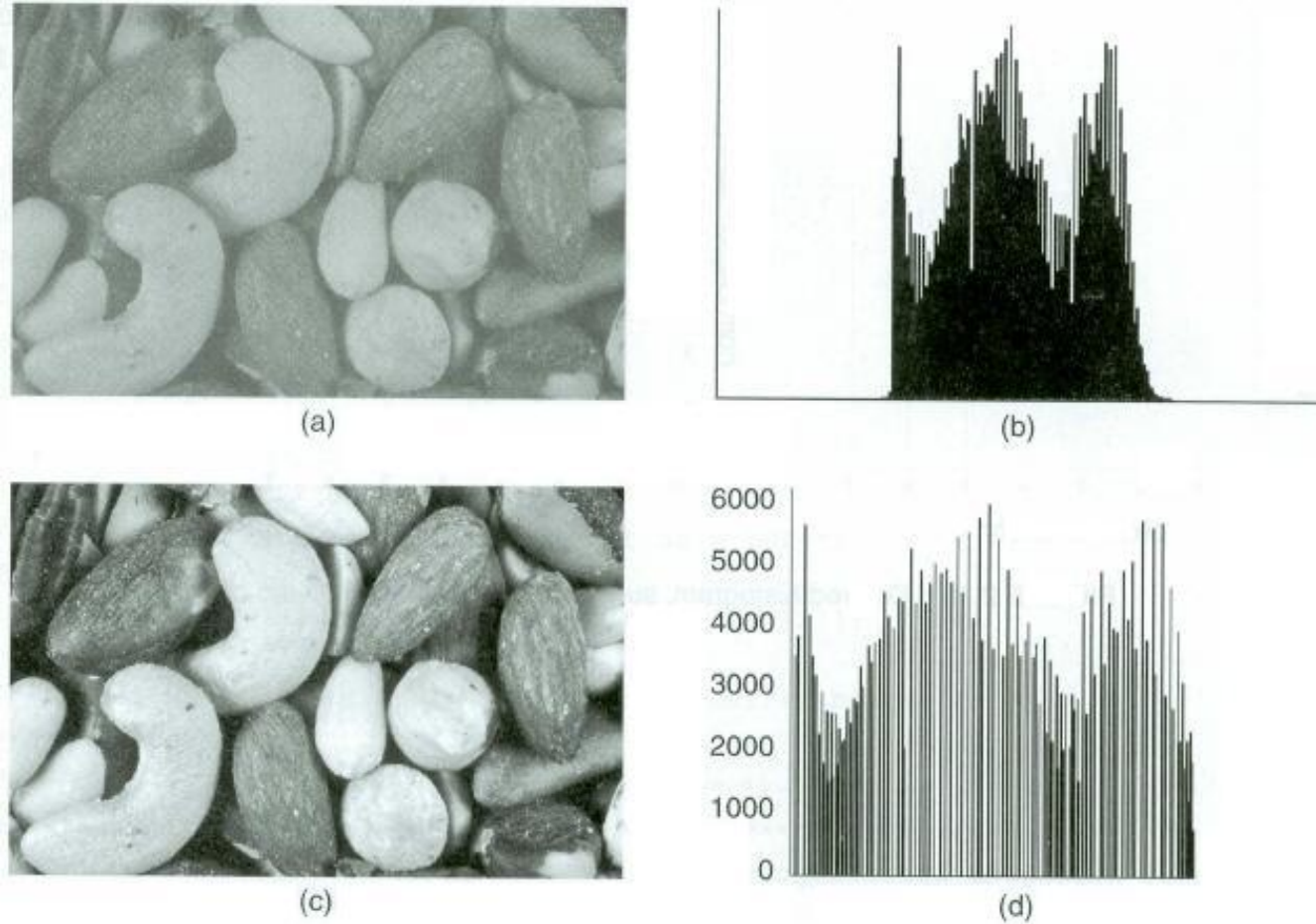
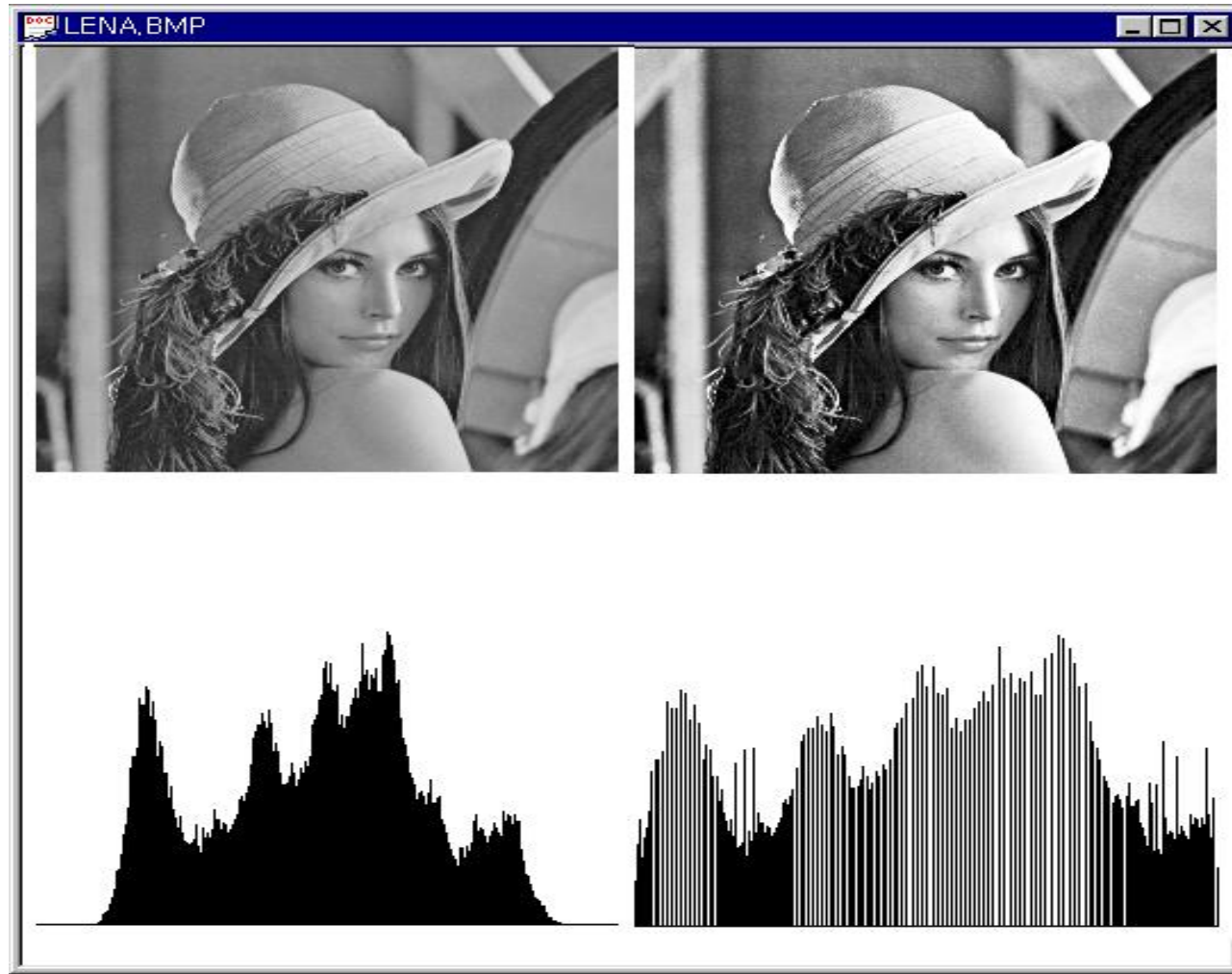


Fig. (a) Original image; (b) histogram of original image; (c) histogram equalized image; (d) histogram of equalized image.

Histogram Equalization



Original Image

Equalization
(256 Level)

Histogram Equalize
Image

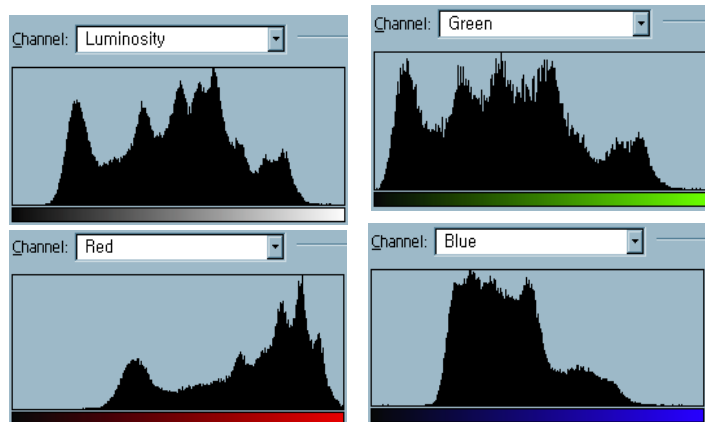
Histogram Equalization



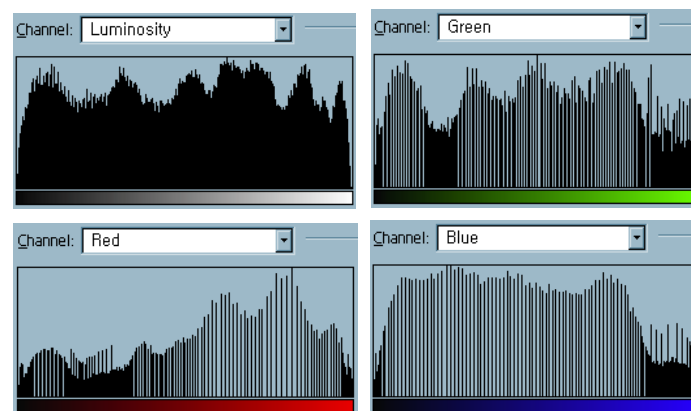
Original Image / 4

Histogram Equalize Image
(64 Level)

Histogram Equalization

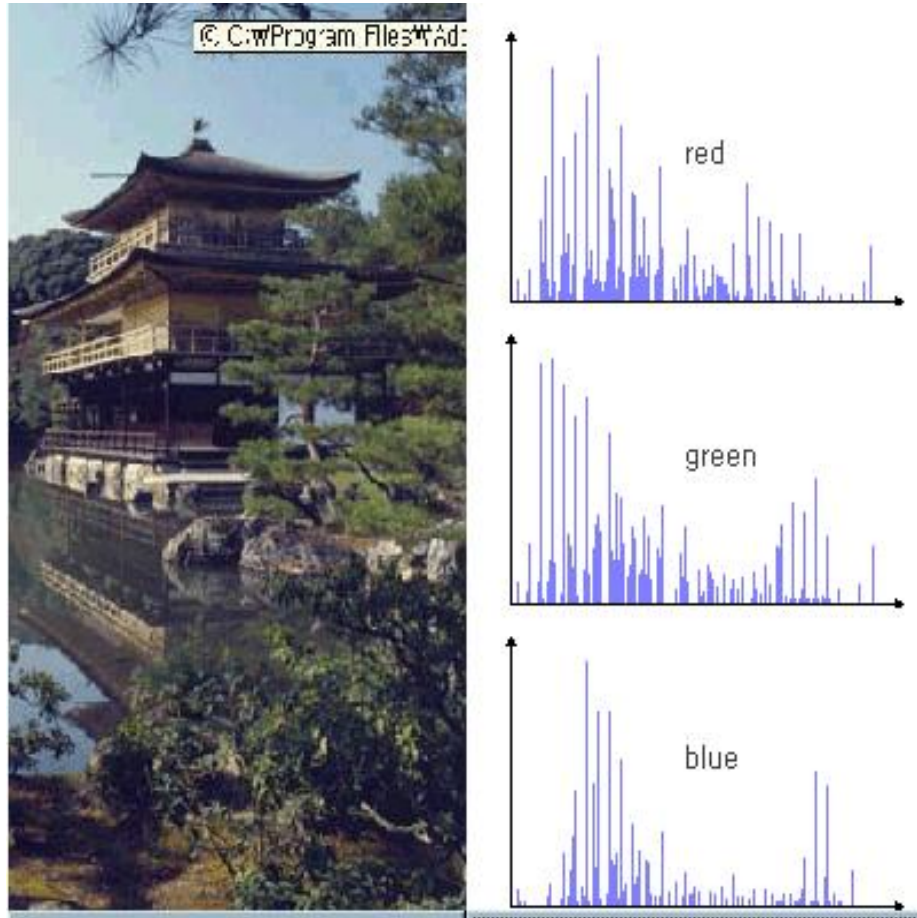


Original Image

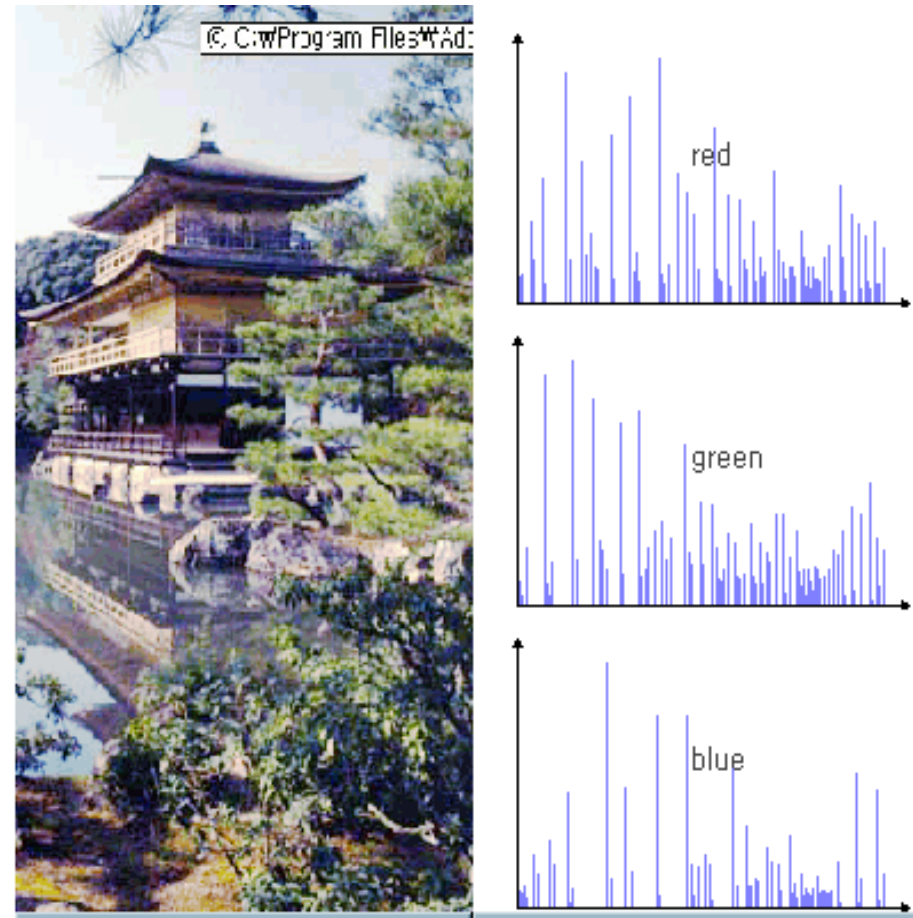


Equalized Image

Histogram Equalization



Original Image



Equalized Image

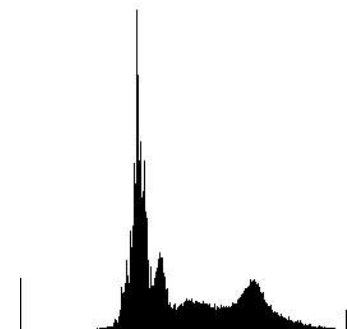
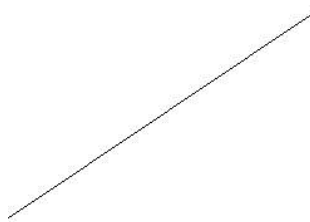
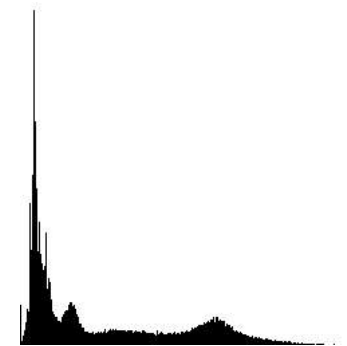
Histogram Equalization

Input / Output

Saturation
adjustment
function

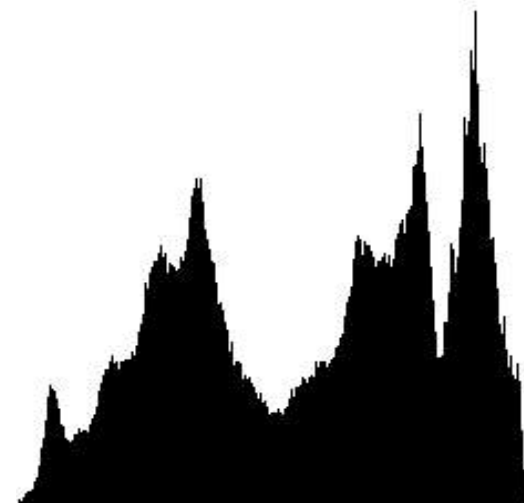
Saturation
image

Saturation
histogram

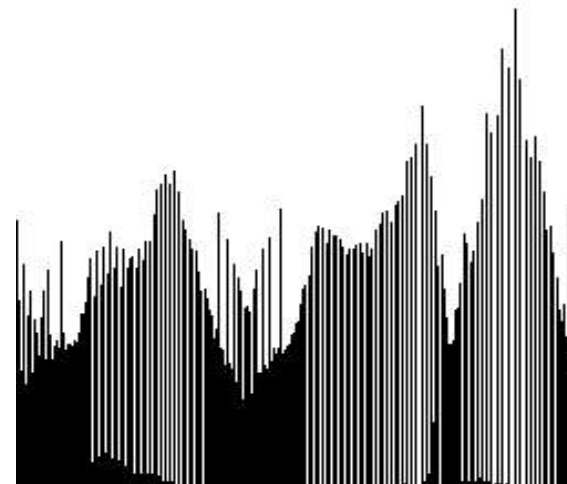


Histogram Equalization

Image



Histogram
equalized
intensity



Histogram Equalization

Original image



Histogram
equalized
intensity



Each R, G, B image
is histogram
equalized



Histogram Specification

- ▶ Histogram specification
 - ▶ Specify particular histogram shapes capable of highlighting certain gray-level ranges in an image
 - ▶ Lighten or darken an image or you need more contrast in an image
 - ▶ Requires both a desired histogram and the image as input.
- ▶ The principle difficulty in applying Histogram Specification lies in being able to construct a meaningful histogram
 - ▶ 1. Specify a particular probability density function.
 - ▶ 2. Specify a particular histogram shape by means of graphic device.

Histogram Specification

- ▶ 1. Compute the histogram of both the input image and the target image.
- ▶ 2. Calculate the cumulative distribution function (CDF) of both the input and target histograms.
- ▶ 3. Find the mapping function that relates the input image's CDF to the target image's CDF.
- ▶ 4. Apply the mapping to the input image to adjust its pixel intensities, resulting in the transformed image.

Histogram Specification

- ▶ Let $p_r(r)$ and $p_z(z)$ be the original and desired probability density functions. Z is the levels of the desired image highlighted certain gray-level ranges

(1) Equalize the levels of the original image

Histogram equalization applying

$$s = T(r) = \int_0^r p_r(w)dw.$$

(2) Specify the desired density function and obtain the transformation function $G(z)$

$$v = G(z) = \int_0^z p_z(w)dw.$$

- ▶ Desired image could be equalized by

(3) Apply the inverse transform function.

$$z = G^{-1}(s) \Rightarrow z = G^{-1}[T(r)]$$

- ▶ The inverse process

Histogram Specification

1. Histogram equalize the original image
2. Perform an inverse histogram equalization on the equalized image

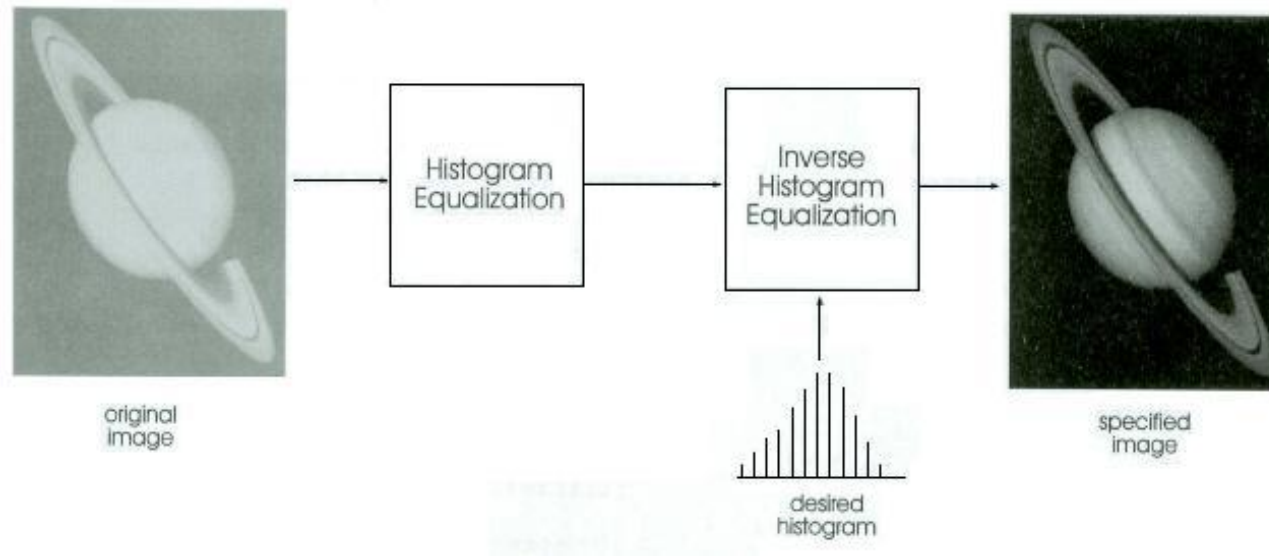


Fig. Block diagram of histogram specification.

Histogram Specification

► Histogram specification

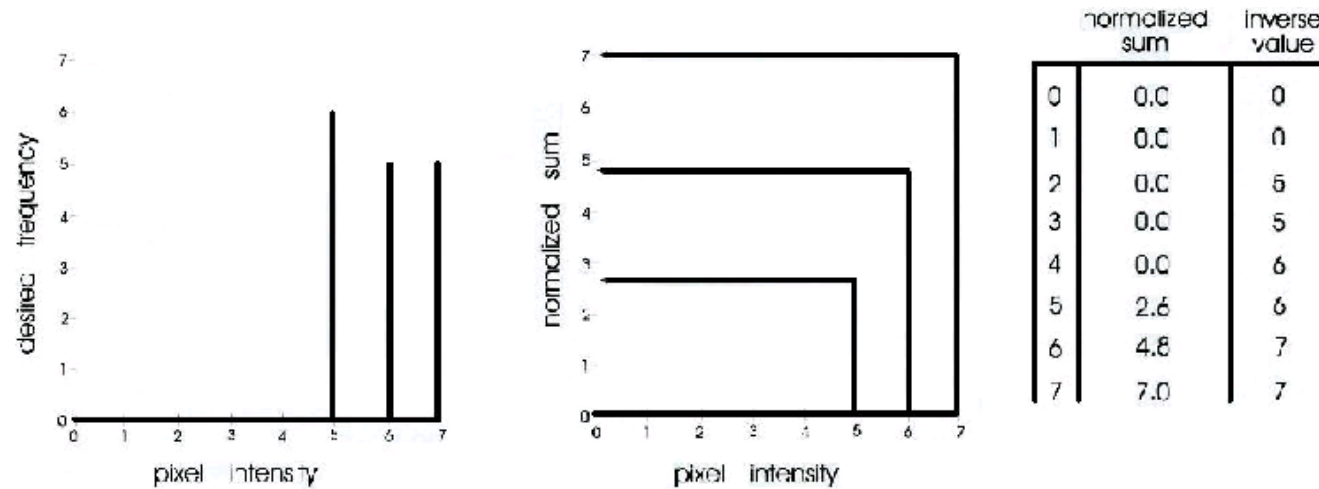
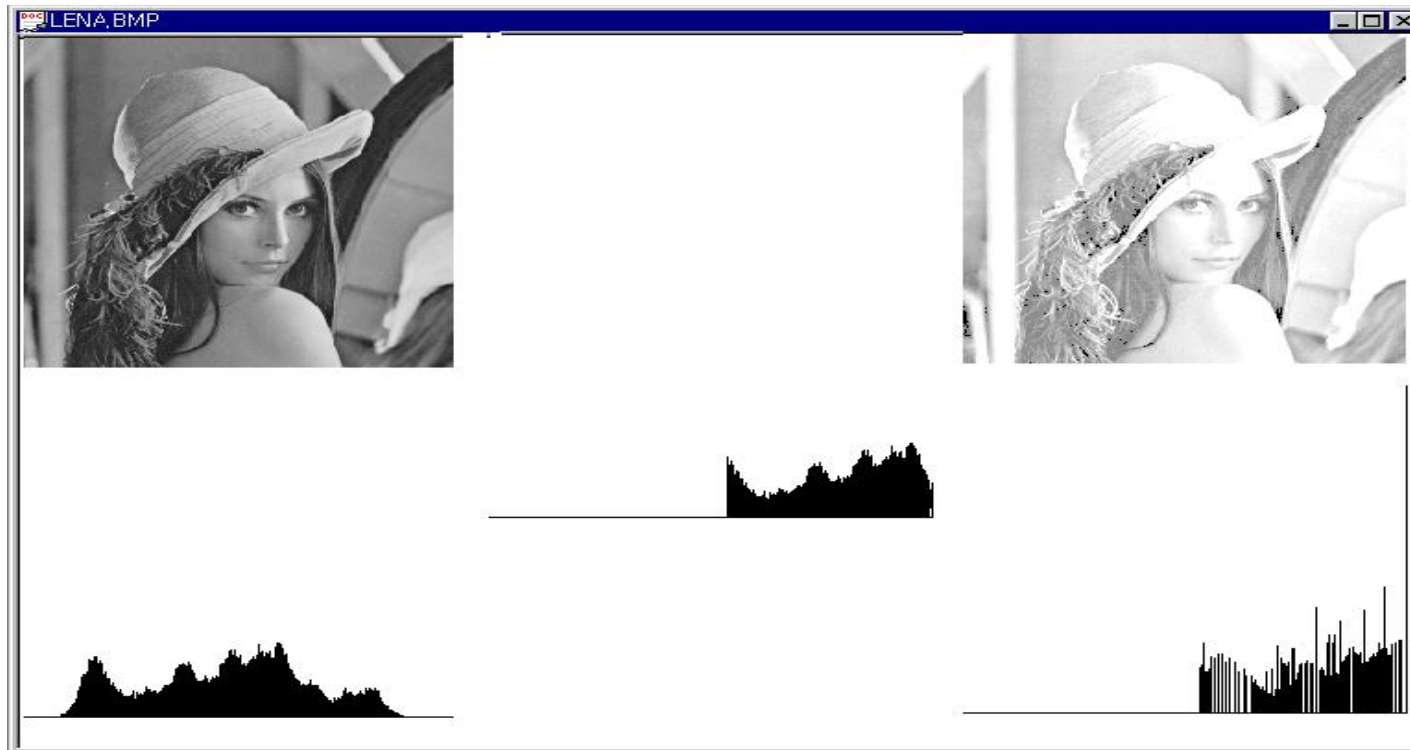


Fig. Desired histogram, sum of histogram with mappings, and inverse LUT.

Histogram Specification



Original Image
& Histogram

Desired Histogram

Result Image
& Histogram

Arithmetic operations

- ▶ Arithmetic operations
 - ▶ Addition, Subtraction, Division, Multiplication
 - ▶ Addition & subtraction: adjusting the brightness of the image

$$I(x, y) + a = I'(x, y)$$

- ▶ a : random pixel value
 - ▶ Subtraction : motion detection
 - ▶ Division, Multiplication : adjust the contrast of the image

Arithmetic operations



(a)



(b)



(c)

Fig. 2.1 (a)Original image; (b)image+40; (c)image-40.



(a)



(b)

Fig. (a)image*1.2; (b)image/1.2.

Arithmetic Operations

- ▶ Addition and Subtraction
 - ▶ Adding values to a pixel brightens the image.
 - ▶ Subtracting values from a pixel darkens the image.
 - ▶ Adding and subtracting cannot adjust the contrast of the image.



<Original>



<Original+50>

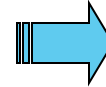


<Original-60>

Arithmetic operations



+

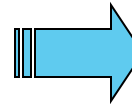


Original image

add operation



-



subtract operation

Arithmetic Operations

- ▶ Multiplication and Division
 - ▶ Multiplying and dividing pixels by different values adjust the contrast of the image.
 - ▶ The contrast of the image will be increased by multiplication and decreased by division.



<Original>



<Original * 1.5>

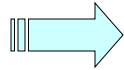


<Original/1.5>

Point processes



Original
image



Multiplication
operation



Division
operation

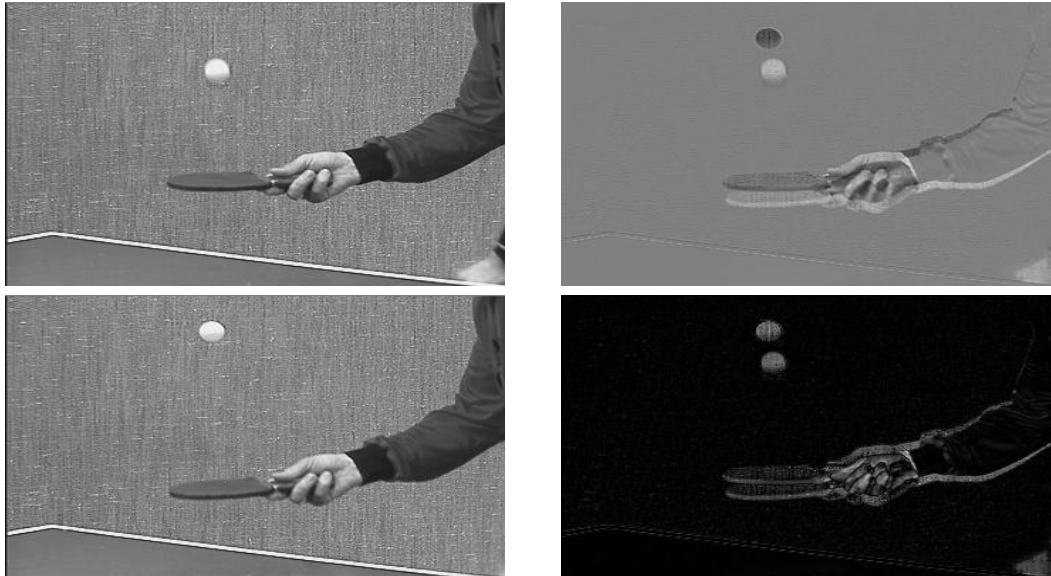
Arithmetic Operations

► Problems

- They can generate negative values or values greater than the maximum possible values.
- One possible solution is clamping. It sets to 0 for negative values and to 255 for overflow values.

Point processes

- ▶ Image Subtraction and Change Detection
 - ▶ applications
 - ▶ medical imaging application : display blood-flow paths
 - ▶ automated inspection of printed circuits
 - ▶ security monitoring



Logic operations

- ▶ NOT operations
- ▶ XOR operations
 - ▶ Bitwise XOR(eXclusive OR) operation
- ▶ AND operations
 - ▶ Bitwise AND operation
- ▶ OR operations
 - ▶ Bitwise OR operation

NOT Operations

- ▶ *new Pixel = 255 – old Pixel*



Original



NOT Operation

XOR Operations

- ▶ XOR function sets bits that are the same in each operand to 0 and bits that are different to 1.
- ▶ In image processing, XOR operation sets the pixels that are specified to black, others to nonblack.
- ▶ Usage
 - ▶ Used to find all pixels of a certain value
 - ▶ Frequently used to draw a cursor for the mouse
 - ▶ For some special effects



XOR Operations



<ORIGINAL>



<ORIGINAL XOR 128>



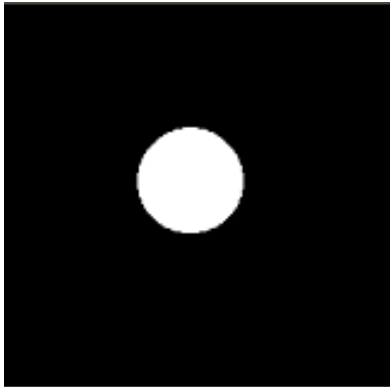
<ORIGINAL XOR 250>

AND operations

$$I_1(x,y) = 10010011$$

$$I_2(x,y) = 01100110$$

$$I_3(x,y) = I_1(x,y) \text{ AND } I_2(x,y) = 00000010$$



AND



=



OR operations

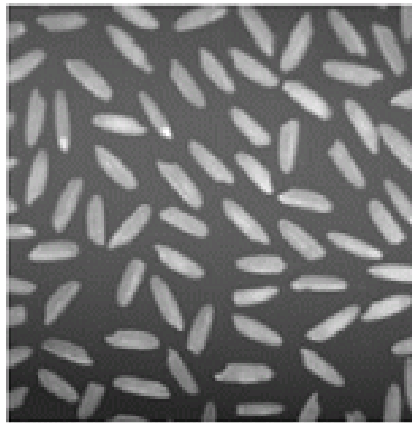
$$I_1(x,y) = 10010011$$

$$I_2(x,y) = 01100110$$

$$I_3(x,y) = I_1(x,y) \text{ OR } I_2(x,y) = 11110111$$



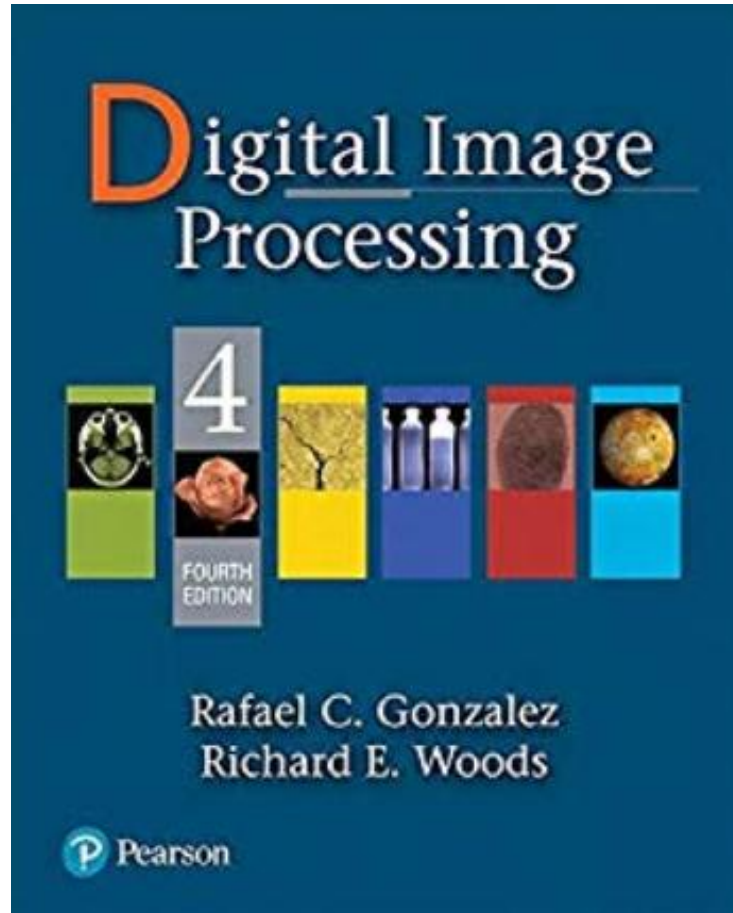
OR



=



Reference



Question



Thank you

